

AMIT KUMAR

M.Sc. Electronics

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OBJECTIVE

M.Sc. Electronics graduate with hands-on MATLAB experience from Mathematical Computation Lab coursework — including ODE/PDE solving, optimization, curve fitting, and statistical analysis — seeking to contribute as a Subject Matter Expert at ORSA Lab. Electronics training brings a strong applied-mathematics and signal-analysis foundation, complemented by basic/intermediate Python and C++ programming and working exposure to SQL. M.Sc. dissertation research applying multiple precision measurement techniques and resolving data discrepancies has built a disciplined, evidence-based analytical approach directly relevant to mathematical model development and systems analysis support.

EDUCATION

M.Sc. Electronics, Department of Electronic Science, University of Delhi South Campus, 2023 – 2025

CGPA: 7.27 / 10 · Relevant Coursework: Semiconductor Devices, VLSI Technology, Digital Circuit Design, Analog Circuit Design, Fabrication & Characterization Techniques, Mathematical Computation Lab (MATLAB)

B.Sc. (Hons.) Electronic Science, Acharya Narendra Dev College, University of Delhi, 2020 – 2023

CGPA: 7.43 / 10 · Key Subjects: Digital Electronics & VHDL, Semiconductor Devices, Microprocessors (8085/8051), Signals & Systems

RELEVANT SKILLS

- **MATLAB & Mathematical Modeling:** Hands-on MATLAB experience from Mathematical Computation Lab coursework — ODE/PDE solving, optimization, curve fitting, and statistics; comfortable using core MATLAB tools for data processing, algorithmic problem-solving, and predictive analysis
- **MS Office & Data Analysis:** Comfortable with MS Excel and the MS Office suite for data organization and analysis; Origin Pro for scientific data analysis and visualization
- **Programming (OOP):** Basic/intermediate C, C++, and Python — object-oriented programming fundamentals from coursework and project use
- **Database & SQL:** Basic working exposure to SQL for data querying and management
- **Analytical & Research Rigor:** M.Sc. dissertation experience applying multiple precision measurement techniques, resolving discrepancies across datasets, and documenting findings systematically for an academic supervisor
- **Additional Tools:** Proteus, Multisim, PSpice — circuit simulation and analysis from coursework

RESEARCH EXPERIENCE — M.SC. DISSERTATION

Thin Film Fabrication & Semiconductor Characterization Jan – May 2025

Department of Electronic Science, UDSC · Supervisor: Prof. Manoj Kumar Khanna

- Fabricated $\text{Fe}_{80}\text{Co}_{20}$ thin-film devices (6–15 nm) via DC sputtering, systematically varying vacuum ($\sim 10^{-6}$ Torr), Ar pressure (3–5 mTorr), and deposition power (50–200 W) — structured, parameter-driven experimental design.
- Applied XRD, VSM, and FMR characterization techniques to extract structural and magnetic parameters, independently learning and operating precision instrumentation with zero prior hands-on experience — demonstrating fast technical ramp-up on unfamiliar tools and methods.
- Analyzed and cross-validated data across multiple measurement techniques to resolve discrepancies and draw consistent, defensible conclusions, then documented and presented findings to an academic supervisor — a disciplined, evidence-based analytical approach directly transferable to mathematical modeling and systems analysis.

WORK EXPERIENCE

Electronics Intern, Sagedel Tech LLP Jul – Dec 2023

- Designed and tested analog/digital circuit modules (BJT, 555 Timer IC, Arduino); performed structured functional verification and data logging on prototype electronics kits — early exposure to disciplined, methodical testing practice.

CERTIFICATIONS

- Advanced Certificate Course – HPC (AI), C-DAC ACTS Delhi (July 2026 batch, in progress) — Linux, C/C++/Python programming, and HPC/parallel computing fundamentals
- VLSI for Beginners — NIELIT Calicut, April 2025 (5-day online certification)
- UGC-NET, Electronic Science (Dec 2025) — Qualified for Ph.D. admission (136/300, 77.77 percentile)